Identify emission-abnormal heavy-duty diesel vehicles via on-board diagnostic urea analysis

ABSTRACT. On-board diagnostic (OBD) systems with remote transmission capabilities are increasingly used to monitor NOx emissions from heavy-duty diesel vehicles (HDDVs) across China. Yet, the full potential of OBD data remains untapped due to data quality limitations. This study introduces a data-driven method to identify urea depletion segments and calculate urea consumption rates using median filtering and a sliding window approach. Among 328 daily-scale vehicles, 23 (7%) exhibited no detectable urea use, suggesting possible tampering. However, due to mixed vehicle types and short observation periods, this indicator alone is insufficient. To improve reliability, we analyze monthly- and annual-scale data from same-model vehicles and propose a baseline-based detection framework. Results show signs of tampering or abnormal use in 1 of 6 annual-scale and 4 of 22 monthly-scale vehicles. The approach is scalable, cost-effective, and suitable for integration into big-data platforms, offering policymakers a practical tool for targeted enforcement and emission reduction.

KEYWORDS: On-board diagnostics, NOx emission, Heavy-duty diesel vehicle, Urea consumption

1. Introduction.

Heavy-duty diesel vehicles (HDDVs) are a major source of ambient NOx emissions(Programme, 2024). In China, HDDVs accounted for 76.1% of NOx emissions from all mobile sources in 2023(Ecology and China, 2023), posing significant risks to the ecosystem(Xu et al., 2019), atmospheric environment(Akimoto and Tanimoto, 2022; Bray et al., 2021), and human health(Hu et al., 2024; Wei et al., 2019). To mitigate NOx emissions from HDDVs, the Chinese government implemented the China VI emission standard, which tightened the NOx emission limits under standard test cycles by 77%(Cui et al., 2018). In addition, the regulation introduced new requirements for remote emission management systems. The on-board diagnostics (OBD) system equipped with a remote monitoring terminal was designed to address this issue, enabling real-time collection of vehicle operation and emission data(Xu et al., 2023). According to the China VI-a standard, newly sold HDDVs must be equipped with an OBD system, with China VI-b further requiring real-time data upload via network connection(Wang et al., 2022).

With the phased implementation of relevant regulations, OBD systems have achieved widespread adoption, uploading a substantial volume of data for remote monitoring platforms. These data have supported a growing number of research. Initial studies primarily examined the accuracy and reliability of OBD data. For instance, Cheng et al. and Zhang et al. connected OBD systems to PEMS devices and reported low average relative errors (within ±22%), high Pearson correlation coefficients (>0.95), and strong overall agreement, confirming the data’s credibility(Cheng et al., 2019; Zhang et al., 2020). Building on this foundation, OBD data have since been applied in various domains, including the identification of high-emitting vehicles(Cao et al., 2025a; Yang et al., 2024b; Zhang et al., 2023a), emission inventory development(Lv et al., 2023), overall emission assessment(Li et al., 2025;

Zhao et al., 2024a; Zhao et al., 2024b), and data-driven information mining(Cao et al., 2025b). Zhang et al. introduced a fuel-window method—analogous to the PEMS-based work window approach—to enable precise identification of high emitters(Zhang et al., 2020). Lv et al. developed a real-time NOx emission inventory method using second-level OBD data(Lv et al., 2023). Zhao et al. analyzed emissions from a large number of vehicles and recommended long-term monitoring protocols to assess compliance throughout a vehicle's service life(Zhao et al., 2024a). In addition, Zhao et al. explored the opportunities and challenges of truck electrification using OBD data(Zhao et al., 2024b). Cao et al. proposed a comprehensive framework encompassing data processing, high-emitting vehicle identification, and diagnosis of primary emission causes(Cao et al., 2025a). They also developed an imputation method to handle missing data caused by signal loss or excessively low coolant temperatures during engine cold starts(Cao et al., 2025b). Although the current research has made notable progress, the full potential of OBD big data remains largely untapped, with a wealth of information yet to be mined through appropriate analytical techniques.

OBD data are both extensive and reflective of real-world driving conditions. Unlike PEMS measurements, they do not require complex and labor-intensive experimental procedures, making them a valuable resource for understanding real-world vehicle operation patterns(Agarwal and Mustafi, 2021; Jiang et al., 2022; Li et al., 2022). However, despite their advantages, large-scale OBD datasets suffer from lower data quality compared to PEMS measurements, often containing substantial noise and a large amount of unusable data(Moradi et al., 2020). To effectively extract meaningful information from such data, appropriate analytical methods are essential. Data-driven approaches, such as machine learning and deep learning, offer one promising pathway(Jia et al., 2025; Moradi et al., 2020; Wei et al., 2022). These models are capable of filtering out a portion of the noise, retaining only the dominant patterns and underlying structures. With the aid of model interpretation techniques, these extracted patterns can be further analyzed and understood(Jia et al., 2024a; Jia et al., 2024b). Nevertheless, data-driven methods have some limitations. They typically require significant computational resources and can be costly to implement. Furthermore, the dominant patterns captured by machine learning models may sometimes be incomplete or biased, potentially leading to misleading interpretations, more conventional, numerically grounded approaches to data analysis are also needed. Such methods are generally more cost-effective and, in many cases, offer sufficient accuracy and robustness for practical applications. Signal denoising is a well-established data processing technique that aims to extract meaningful information from raw signals by suppressing or eliminating noise that may obscure underlying patterns. It has been widely applied in various fields, including vehicle trajectory analysis and the removal of noise from emission measurements(Gao et al., 2024; Li et al., 2021; Sun, 2024). For OBD datasets with high levels of noise, combining signal denoising with indicator-based computation may offer a feasible analytical pathway. Once the noise is effectively filtered out, certain latent information embedded in the data may become accessible.

Urea consumption rate is one of the key pieces of information embedded within raw OBD data. It can be derived by applying signal denoising techniques to parameters such as urea level, NOx concentration, and intake air flow. As a core indicator of the operational status of the SCR system, the urea consumption rate can assist enforcement officers and policymakers in efficiently assessing vehicle conditions and holds potential for identifying tampered vehicles. Urea is the primary reductant used in the SCR systems of HDDVs, Within the SCR system, urea reacts with NOx to produce N_2 and H_2O(Herrero and Ullah, 2020). When functioning properly, the SCR system can significantly reduce NOx emissions(Shiyu et al., 2022; Zhang et al., 2025; Zheng et al., 2022). However, if the urea injection system is tampered with or disabled, the SCR system becomes inoperative, leading to emissions that may be several times—or even dozens of times—higher than normal levels(Li et al., 2022). Some reports have indicated that certain drivers may physically interfere with the urea injection process (Zhang et al., 2023b), slowing it down or halting it entirely. As a result, these HDDVs consume little to no urea and become so-called "super emitters"(Yang et al., 2024a). A common feature of such vehicles is abnormally low urea consumption rates, as the primary motivation behind such modifications is to reduce urea usage in order to cut operating costs. Therefore, a method can be developed to calculate urea consumption rates and screen for vehicles with abnormal consumption behavior.

In earlier studies, Zhang et al. explored this possibility and concluded that the decrease in urea level is not a sensitive indicator of NOx emission control performance(Zhang et al., 2020). Their findings were robust and reasonable given the data conditions and pilot context at the time—especially considering that the OBD system was still under refinement and data resources remained limited. However, several years have passed since that early study, during which the OBD system has been widely deployed, and data availability is no longer a major constraint for investigating this relationship. Despite this progress, the topic has received little further attention in recent years, highlighting the need for a renewed investigation into its potential. Building on the above context, this study proposes a method for estimating urea consumption rates based on signal denoising and indicator calculation, using daily-level OBD data from 432 HDDVs collected via Tianjin Ecology and Environment Bureau's OBD Platform. To investigate the baseline urea consumption of vehicles of the same model, data from an additional 28 vehicles were also collected. The main contributions of this study are as follows: (1) A method was established to identify urea consumption segments and calculate urea consumption rates; (2) Fleet-level urea consumption rates were quantified across different vehicle models; (3) The feasibility of identifying cheating behaviors or retrofitted vehicles was further explored based on urea consumption rate and emission factors (EFs) for a given vehicle model. The findings offer practical insights for policymakers and support the refinement and optimization of remote supervision strategies.

2. Material and Method.

2.1. Data Acquisition and Processing.

The data used in this study were obtained from the Tianjin Ecology and

Environment Bureau's OBD Platform. Specifically, data from 432 vehicles covering one day, 22 vehicles covering one month, and 6 vehicles covering one year were collected. Raw OBD data often contain lots of errors, necessitating thorough data cleaning. Following previously procedures, we applied the same data cleaning method(Cao et al., 2025b), which is described again here for clarity.

The data cleaning process began with addressing timestamp errors, which included discontinuities and duplicates. Timestamp discontinuities were not corrected during cleaning but retained for use in subsequent steps. For duplicated timestamps, two scenarios were distinguished: if the duplicated rows were identical across all columns, the redundant rows were simply removed; if the rows had the same timestamp but differing data, we checked whether a timestamp discontinuity segment of matching length followed immediately. If so, the missing data were filled with these values; if not, the affected segments were resampled. Next, out-of-bounds errors caused by weak signals, NOx sensor shutdown at low coolant temperatures, were identified and removed. Any row containing such values was discarded entirely to ensure data reliability. Finally, constant value errors were addressed, where sensor outputs remained fixed over time, often due to malfunction returning default values—for example, some HDDVs repeatedly report a speed of 65.535 km/h, a value that is physically implausible under real-world driving conditions. To detect this, a threshold of 15 consecutive identical values (excluding zero speed) was applied, and data exceeding this threshold were removed. Representative examples for each category of error are shown in Figure S1.

2.2 Identification of Valid Urea Consumption Segments.

2.2.1 Categorizing Urea Level Depletion Types.

After data cleaning, the remaining data are generally reliable. An examination revealed that urea sensors typically fall into two categories: 1. Sensitive type. These sensors respond strongly to external noise such as vehicle bumpiness. Locally, the readings fluctuate frequently, making short-term segment-based urea consumption calculations unreliable. However, from a broader perspective, the signal tends to show a clear downward trend, and long-term segments can offer stable and representative estimates. 2. Threshold-triggered type. These sensors exhibit stepwise drops in urea level. Instead of a gradual decline, the level remains constant until a relatively large threshold (much greater than the sensor resolution) is crossed, after which a rapid and substantial drop occurs within a short period. This drop may appear as a sharp vertical drop, a brief continuous decrease, or sustained fluctuations between two distinct levels. An algorithm can be developed to automatically identify such drop segments. Representative examples of both types are shown in Figure S2.

Different segmentation algorithms are required for different sensor types. Therefore, classifying the sensor type is critical. We propose a method that performs well on our data and may be of practical use for large-scale classification on OBD platforms. The procedure is as follows:

1. Apply a median filter to the urea level signal for noise reduction. The median filter is a non-linear digital filtering technique commonly used to remove noise from a signal or image. It is especially effective at preserving edges while reducing impulsive

noise(Gupta, 2011; Justusson, 2006). Let the original urea consumption signal be u_i ($u_i = [u_1, u_2, \dots, u_n]$), the median filter replaces each point with the median value of the points within a specified kernel centered around it. The window size is typically an odd number k to ensure a central element. Formally, the filtered signal $\hat{u_i}$ is defined as:

$$\hat{u_i} = median(u_{i-r}, u_{i-r+1}, \dots, u_i, \dots, u_{i+r})$$

Where $r = \frac{k-1}{2}$, and k is the kernel size, which is 299 in the study. Although this value may appear large and raise concerns, the urea consumption rate is very slow, and long-term drifting segments can occur. Smaller kernel sizes fail to suppress these, whereas a larger size effectively removes such noise. Furthermore, the filtered data have minimal influence on downstream classification or calculation steps. To justify this choice, additional discussion is provided later.

2. Compute two statistical parameters based on the filtered data. Value 1 is the average duration of segments where the urea level remains unchanged. Value 2 is the maximum duration of a segment with constant urea level. The computation steps are as follows:

First, compute the first-order difference:

$$\Delta\hat{u_i} = \hat{u_i} - \widehat{u_{i-1}}$$

When Δ_i, it indicates the current value equals the previous one. Segments are numbered using:

$$s_i = \sum_{j=1}^{i} \mathbb{I}\left(\Delta_j \neq 0\right)$$

Where $\mathbb{I}(\cdot)$ is the indicator function. Let T_k be the length of the $k-th$ segment.

Then,

$$\text{value1} = \frac{1}{m}\sum_{k=1}^{M} T_k \cdot \mathbb{I}\,(T_k > 100)$$

$$\text{value2} = \max\{\, T_k \mid \widetilde{y_k} < 100$$

The thresholds in value1 and value2 deserve explanation. In value1, only segments longer than 100 are considered, since short, fragmented drops in threshold-triggered sensors could otherwise skew the average downward and lead to misclassification. For the calculation of value2, segments with a urea level equal to 100 are excluded. This is because investigations have shown that some vehicles display a persistent reading of 100 while the actual urea level has already exceeded the sensor's maximum scale. Such readings result in abnormally slow apparent consumption rates. If not removed, these segments would artificially inflate value2 and could lead to misclassification of certain sensitive-type faults as threshold-triggered.

3. If a vehicle satisfies both value1<900 and value2<4000, it is classified as sensitive; otherwise, as threshold-triggered.

2.2.2 Identification of Urea Consumption Segments.

Different segmentation strategies are applied depending on the sensor type. For threshold-triggered sensors, we first define a "urea consumption event" as the process from one reliable reading to the next. When the level begins fluctuating after a stable period, it indicates the true value is transitioning between two stable points, and the readings are unreliable. Once a new stable value is reached, the reading is considered reliable again. The difference between two reliable points represents the amount of urea consumed. The identification algorithm is as follows:

1. Apply a median filter with kernel size 299 to the raw urea sequence u_i, yielding $\widehat{u_i}$.

2. A moving average with a window size of 500 is applied to the median-filtered sequence $\widehat{u_i}$. While moving averages are commonly used for curve smoothing, here they serve as a detector. When the window moves, if the values remain stable, the corresponding average remains unchanged; conversely, if the values change, the average also shifts. This property of moving averages enables the identification of stable segments longer than 1000 in length. The resulting smoothed sequence is denoted as $\widetilde{u_i}$, given by:

$$\widetilde{u_i} = \frac{1}{1000} \sum_{j=i}^{i+999} \widehat{u_j}$$

3. Compute the first-order difference:

$$\Delta\widetilde{u_i} = \widetilde{u_i} - \widetilde{u_{i-1}}$$

Then number each segment:

$$s_i = \sum_{j=1}^{i} \mathrm{II}\left(\Delta\widetilde{u_i} \neq 0\right)$$

Define the segments as:

$$\mathrm{segment}_k = \{i \mid s_i = k\}$$

4. The following criteria are used to identify usable segments: (1) the segment length must exceed 1000; (2) if a temporal discontinuity is present, the proportion of valid data duration to the total duration must exceed 0.8; (3) the segment must not contain any urea refilling events. Given that most refilling behaviors involve a substantial increase from a lower to a higher level, and that the filtered data typically

do not exhibit large fluctuations, a urea refilling event is simply defined as any increase exceeding 50% within the segment; and (4) for daily-scale data, the first segment of each day is excluded, as it is unclear whether the vehicle had already been in operation or was just starting up. Segments meeting all these criteria are considered valid for subsequent calculations.

For sensitive-type vehicles, although readings are accurate, the noise level remains high. Hence, median filtering is still required. Let the filtered sequence be $\widehat{u_i}$, and compute first-order differences $\Delta\widehat{u_i}$. Number the resulting steps using s_i, and define:

$$\mathrm{step_k} = \{i \mid s_i = k\}$$

The length of each step stage, denoted as L_k, is calculated. Consecutive step stages are merged if their cumulative length satisfies the condition:

$$\sum_{p=k}^{k+r} L_p > 2000$$

In this case, the $r + 1$ adjacent step stages are combined into a single complete segment. This process continues until the entire dataset is fully segmented. For these segments, we applied the same segment filtering method as used in the threshold-triggered approach to retain valid segments.

2.3 Indicator Calculation.

To assess overall and segment-specific NOx emissions, we define several indicators. The instantaneous downstream NOx mass emission rate at time t, $\mathrm{NO_{x,down,t}^{mass}}$ (g/s), is calculated as:

$$\mathrm{NO_{x,down,t}^{mass}} = \frac{0.001 \times 46}{3600 \times 22.4} \times \frac{1}{\rho_e} \times NO_{x,\text{down-stream},t} \times (FuelFlow_t \times \rho_{\text{fuel}} + MAF)$$

Where $\rho_e = 1.29 kg/m^3$ (air density), $NO_{x,\text{down-stream},t}$ is the downstream NOx concentration in ppm, $FuelFlow_t$ is the fuel flow in g/h, $\rho_{\text{fuel}} = 0.85 kg/L$, and MAF is the mass air flow in kg/h. For a segment of length T, the fuel-based NOx EF (g/kg-fuel) is given by:

$$NOx_{\text{fuel-based}} = \frac{3600 \times \sum_{t=1}^{T} \mathrm{NO}_{\mathrm{x,down,t}}^{\mathrm{mass}}}{\rho_{\text{fuel}} \times \sum_{t=1}^{T} FuelFlow_t}$$

This indicator is conceptually similar to power-based EFs, as it reflects the level of NOx emissions. However, since the net output torque uploaded by the OBD system is expressed as a percentage relative to its maximum value, it is not feasible to accurately calculate engine power. Therefore, a fuel-based EF is used instead. The upstream instantaneous NOx mass emission rate $\mathrm{NO}_{\mathrm{x,up,t}}^{\mathrm{mass}}$ (g/s):

$$\mathrm{NO}_{\mathrm{x,up,t}}^{\mathrm{mass}} = \frac{0.001 \times 46}{3600 \times 22.4} \times \frac{1}{\rho_e} \times NO_{x,\text{up-stream},t} \times (FuelFlow_t \times \rho_{\text{fuel}} + MAF)$$

Where $NO_{x,\text{up-stream},t}$ is the upstream NOx concentration in ppm. This indicator represents the instantaneous mass of NOx emitted from the engine. The total NOx consumption $\mathrm{NO}_{\mathrm{x,SCR\text{-}consumed}}^{\mathrm{mass}}$ (g) by the SCR system is defined as:

$$\mathrm{NO}_{\mathrm{x,SCR\text{-}consumed}}^{\mathrm{mass}} = \sum_{\mathrm{t=t_0}}^{\mathrm{t_1}} \mathrm{NO}_{\mathrm{x,up,t}}^{\mathrm{mass}} - \sum_{\mathrm{t=t_0}}^{\mathrm{t_1}} \mathrm{NO}_{\mathrm{x,out,t}}^{\mathrm{mass}}$$

The NOx consumption factor (CF), denoted as $NOx_{\text{fuel-based,SCR-consumed}}$ (g/kg-fuel), is calculated as:

$$NOx_{\text{fuel-based,SCR-consumed}} = \frac{3600 \times \mathrm{NO}_{\mathrm{x,SCR\text{-}consumed}}^{\mathrm{mass}}}{\rho_{\text{fuel}} \times \sum_{t=1}^{T} FuelFlow_t}$$

To some extent, it reflects the efficiency of the SCR system. Moreover, the sum of this indicator and $NOx_{\text{fuel-based}}$ corresponds to the upstream EF. For a given urea consumption segment with a valid duration of T, the urea consumption rate

(%urea/kg-fuel) can be calculated as:

$$\mathrm{Urea}_{\text{fuel-based}} = \frac{3600 \times \Delta\hat{u}}{\rho_{\text{fuel}} \times \sum_{t=1}^{T} \mathrm{FuelFlow}_{\mathrm{t}}}$$

Where $\Delta\hat{u}$ is urea level decrease during the smoothed segment.

$$\Delta\hat{u} = \max(\hat{u}) - \min(\hat{u})$$

Here, the denominator—fuel consumption—serves to quantify the vehicle's driving behavior.

3. Results and Discussion.

3.1 Data Processing and Emissions Analysis.

Data processing was executed as the initial step. Figure 1 presents the proportion of remaining data, and the shares of data removed due to constant-value errors, out-of-bounds errors, and timestamp errors across 460 vehicles. From the figure, constant-value errors are relatively low, with a median 0.74%. Out-of-bounds errors are more frequent, yet still remain mostly below 25%, with a median around 10%. Timestamp errors are exceptionally high, with a median close to 50% and most values clustering near this level. Such a high incidence of timestamp errors substantially lowers the rate of remaining usable data (median ~30%). This result is somewhat surprising, as studies by Zhao et al. and Lv et al. suggest timestamp errors should be much less frequent, with a higher proportion of valid data(Lv et al., 2023; Zhao et al., 2024a). Our empirical review revealed that while most vehicles recorded data within the expected ~24-hour period (86,400 s), 110 vehicles substantially exceeded this duration—notably including a maximum recording span of 205,727 s—suggesting systemic data duplication issues. To confirm this, we removed timestamp-error records and recalculated the other error rates, as shown in Figure S3. The process raises the median usable-data rate to approximately 70%. Since timestamp duplication does not entail information loss, the cleaned dataset remains suitable for analysis. Nonetheless, incorporating automated deduplication on data platforms is recommended to conserve resources. Notably, out of 460 vehicles, 104 had no usable data at all, likely due to sensor failure or possibly tampering—further investigation is needed. Consequently, our subsequent analysis focuses on the remaining 356 vehicles, with 328 on daily and 28 on monthly/annual scale.

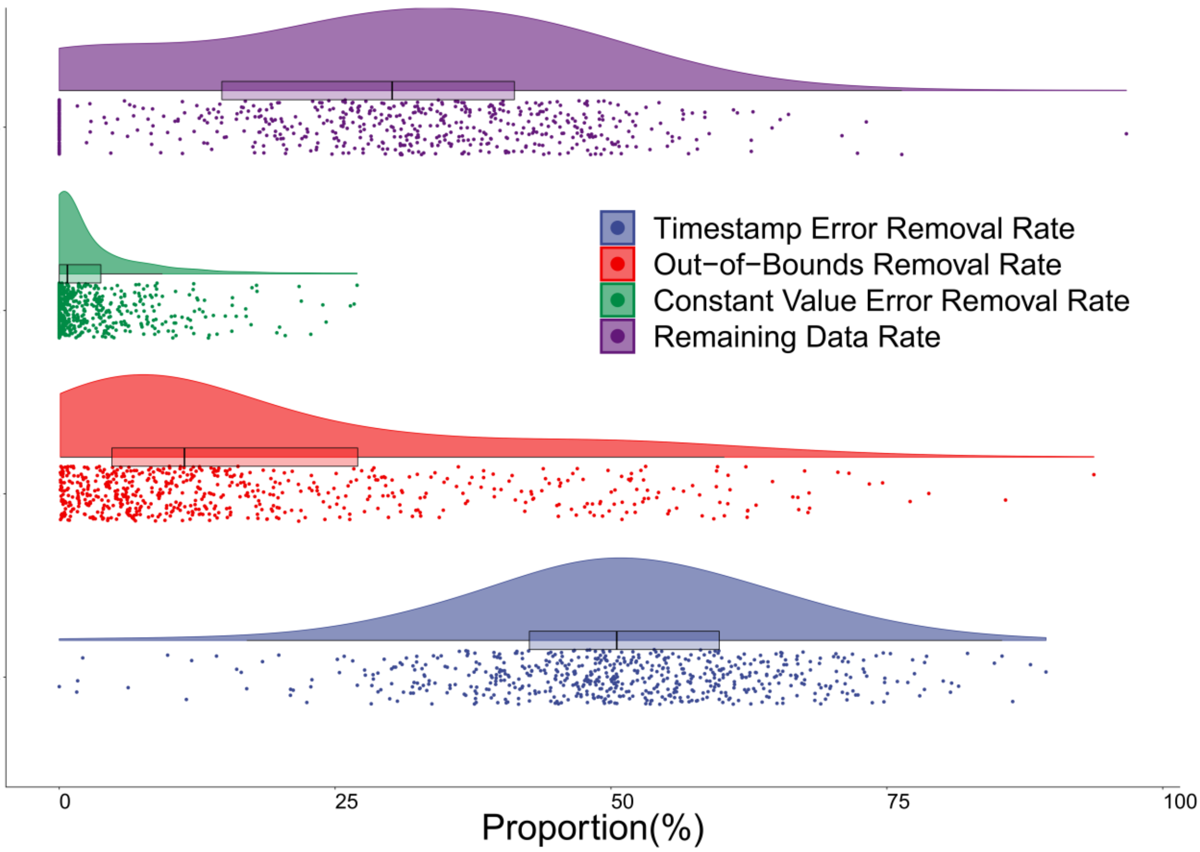

Figure 1. Proportional Distribution of Different Error Removal and Data Retention Rates in OBD Datasets.

Based on the processed daily-scale data, figure 2 was generated, where the x-axis represents the fuel-based NOx EF, and the y-axis indicates the fuel-based NOx CF, the mass of NOx removed by the SCR system. Each point corresponds to one vehicle, with its size representing the volume of available data. The green and blue lines mark the PEMS in-use limit (2.9 g/kg-fuel) and the Diesel Engine Limit (1.94 g/kg-fuel)(Zhao et al., 2024a), respectively. The color intensity reflects the sum of downstream NOx emissions and SCR-consumed NOx, which approximates the engine-out NOx EF. Among the vehicles analyzed, 47% exceed the PEMS limit, and 58% exceed the engine-out limit. The result does not directly correspond to the proportion of high-emitting vehicles in the fleet, as proper classification requires the fuel window method and sufficient data length. Nevertheless, they point to a non-negligible risk of excessive emissions among the sampled vehicles, as our previous study suggests a strong correlation between high-emitting vehicles identified through the fuel-window method and those with EFs exceeding regulatory limits(Cao et al., 2025a). These findings underscore the need for further investigation into the overall emission profile in this region. A clear negative correlation is observed between the NOx EF and the NOx CF. This is consistent with the expected behavior of SCR systems: higher SCR efficiency corresponds to lower tailpipe emissions and thus greater NOx consumption. The plot also reveals that vehicles with low engine-out emissions—potentially due to limited operational activity—and sparse valid data may yield underestimated downstream NOx values. Accordingly, in the case of daily-level data, the NOx CF alone is insufficient to characterize vehicle emission performance and should be interpreted in conjunction with engine-out emissions and data completeness.

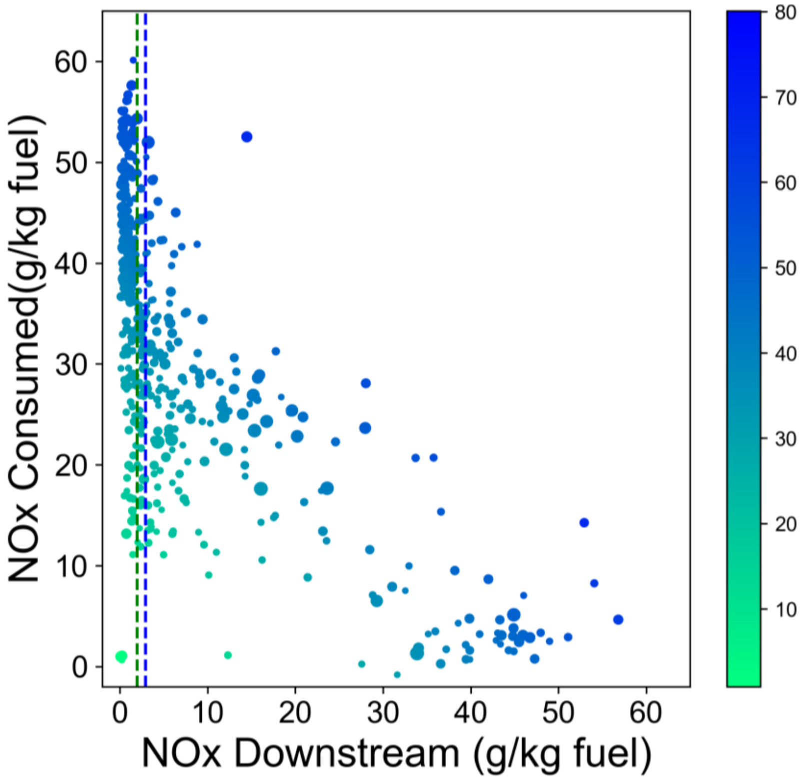

Figure 2. Emission factors and consumption factors of daily fleet-level data. Point size represents the data volume, while color intensity indicates the sum of consumption and emission factors. The green line denotes the Diesel Engine Limits (1.94 g/kg-fuel) and the blue line denotes PEMS in-use limits (2.9 g/kg-fuel).

3.2 Fleet-Level Analysis of Urea Consumption.

The raw data exhibited pronounced noise, necessitating signal denoising to extract meaningful information. We first examine the effect of varying kernel size values on the denoising outcome. Three representative vehicles were selected for illustration. Figures 3a and 3b depict two types of threshold-triggered types. In Figure 3a, sharp, short-term fluctuations are observed just before the urea level drops—they do not reflect actual urea consumption and are considered noise that must be removed. In contrast, Figure 3b shows a relatively stable urea level both before and after a distinct stepwise decline; though generally less noisy, minor fluctuations still occur during the drop. Figure 3c presents a typical sensitive type, with significant noise in the raw data likely caused by vehicle vibration or road conditions, which must be filtered to reveal the primary decreasing trend. Initially, we applied a small kernel size of 5. This setting slightly smooths the curves in all three cases, but substantial fluctuations remain. This is unacceptable—residual noise will prevent the descent segment extraction algorithm from identifying complete segments, which in turn will significantly compromise the accuracy of the calculated urea consumption rate.. Increasing the kernel size to 99 improves the results considerably, yet some residual oscillations remain in Figure 3a), and visible plateaus persist in Figure 3c). Ultimately, a larger kernel size of 299 is chosen, which effectively preserves the main trends while eliminating most of the noise in all three scenarios. It is undeniable that further increasing the kernel size could help eliminate more noise, including some erratic records lasting over 1000 seconds. However, this would also distort the original data to some extent, particularly within the first kernel size seconds. Using a smaller kernel size may result in some residual errors, as their durations are generally short and can be effectively removed by the filtering algorithm. Therefore, as a conservative approach, we selected a kernel size of 299, striking a balance between effective noise reduction and the preservation of original signal characteristics.

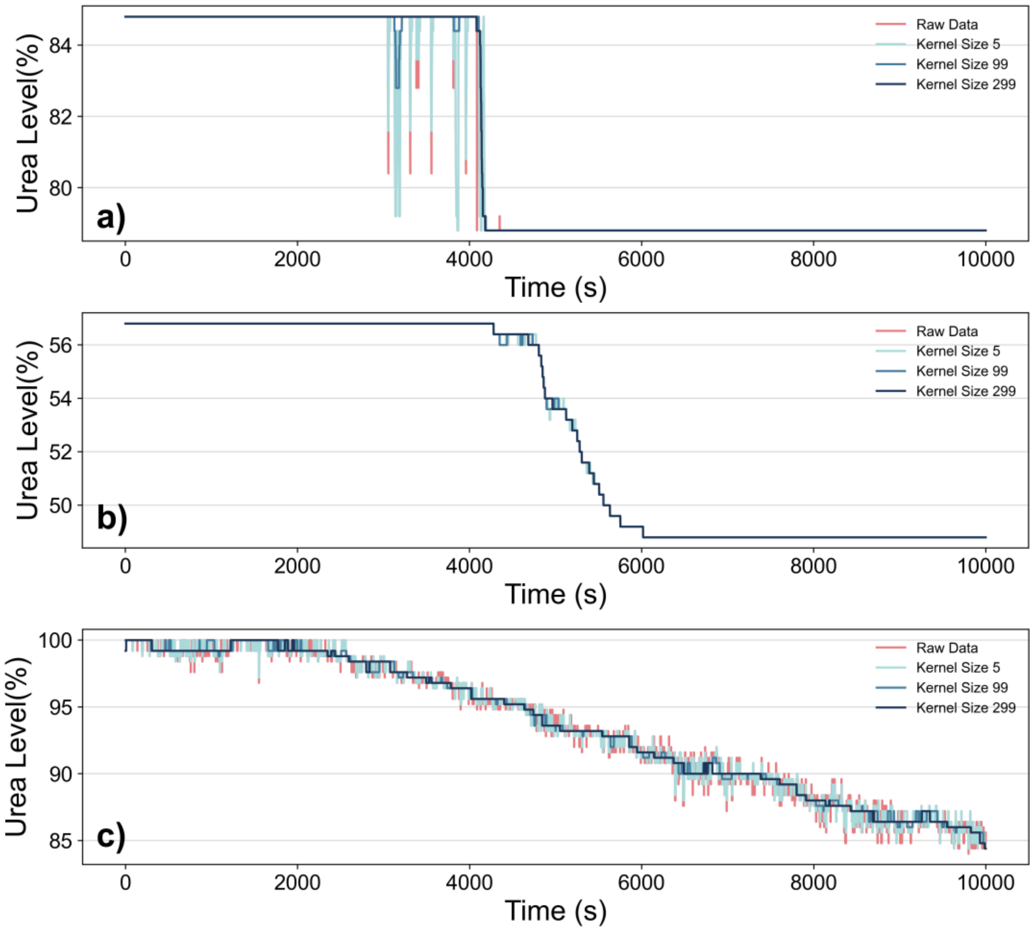

Figure 3. Performance of Medium Filters with Varying Kernel Sizes across Different Types of Urea Level Decline. a) Threshold-triggered type with fluctuations around two discrete levels; b) Threshold-triggered type with a gradual decline after reaching a certain threshold; c) Sensitive type with continuous and smooth decline behavior.

We applied the same denoising process to all daily-scale vehicles. As current literature lacks systematic studies on urea consumption patterns, we visualized the signal-processed urea level variations for all daily-scale vehicles with in Figure 4. Notably, because the raw data vary in length and exhibit low information density—with changes occurring only every few hundred or even thousand data points—direct plotting would be both computationally slow and visually unclear. To address this, we employed Python's lttb package to downsample each time series to 100 points before plotting. As shown in the figure, most vehicles exhibit a healthy downward trend in urea levels. For threshold-triggered vehicles, urea depletion tends to occur abruptly, whereas for sensitive-type vehicles, the depletion is more gradual and continuous. In addition, several urea refilling events can be observed, characterized by a sudden and substantial increase in urea level—typically from a relatively low value to a much higher one, with refill volumes exceeding 50%. Due to data limitations, this study does not explore refilling behaviors in depth; however, future work may consider analyzing refilling frequency as a potential source of additional information. From our calculation, we discovered that the majority of vehicles (293) are threshold-triggered types, while 11 are the sensitive types. A further 24 vehicles show no detectable change in urea level over the observed period. For these vehicles, we plotted the cumulative consumed NOx mass against time in Figure S3. Most vehicles display a steady increase in NOx consumption over time. With the exception of vehicle b), all had EFs exceeding both the PEMS and engine-specific standards. Vehicle b), however, had a relatively short operating duration (10,000 seconds), a low EF (1.3 g/kg-fuel), and a small total NOx mass (150 g), suggesting that it operated under low-load conditions. Based on the available data, its true status remains uncertain. Possible explanations include: (1) the vehicle is genuinely low-emitting, but its urea sensor is faulty or the urea level has not yet reached the threshold that triggers injection (i.e., threshold-triggered type); or (2) the vehicle is actually high-emitting, but due to sustained low-load operation, its emissions appear deceptively low. Except for vehicle b), the remaining vehicles—characterized by high emission factors (mean: 34.2 g/kg-fuel), extended operation durations (>10,000 s), and extremely low NOx conversion factors (mean: 6.61 g/kg-fuel)—are suspected of potential tampering. Previous studies and official reports suggest that, in efforts to reduce urea use and operational costs, some drivers may disable urea sensors or the injection system. While these findings do not constitute definitive evidence of tampering, they highlight vehicles that merit closer inspection and may serve as a future reference point for fleet-wide compliance checks.

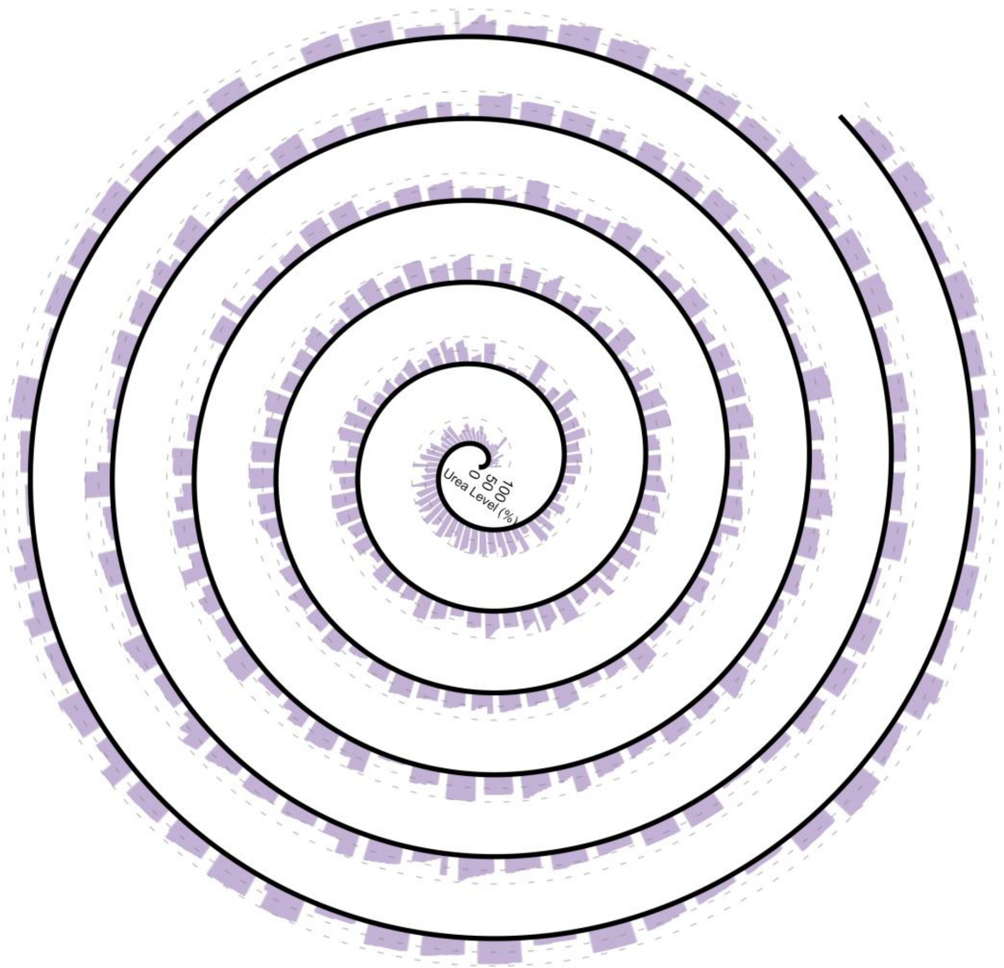

Figure 4. Daily-scale Urea Consumption Plots for all Vehicles. Time Progresses in a Counterclockwise Direction.

Subsequently, we applied the segmentation algorithm described in Section 2.2 to all daily-scale vehicles. Figure 5 provides an example application of the algorithm. After signal denoising, the original noisy series becomes amenable to accurate segmentation. Based on the filtered signals, distinct urea depletion segments were identified, allowing for reliable calculation of urea consumption. However, because the algorithm emphasizes change rather than decline, some urea refilling events and residual noise are mistakenly included as candidate segments, later removed by the filtering process. Overall, the algorithm successfully captures the majority of depletion segments and supports the computation of urea consumption rates. Even if some residual, unfiltered errors remain, their impact can be mitigated by the identification of multiple valid segments, thereby avoiding significant misjudgment. Nevertheless, some limitations remain. Specifically, the window length of 500 s is suitable for most vehicles, but some depletion events exceed this duration, leading the algorithm to incorrectly split a single event into multiple fragments—resulting in skewed consumption rates. To address this, we manually removed such segments; while infrequent in our sample, we recommend developing adaptive algorithms for large-scale applications to automatically identify and exclude such artifacts.

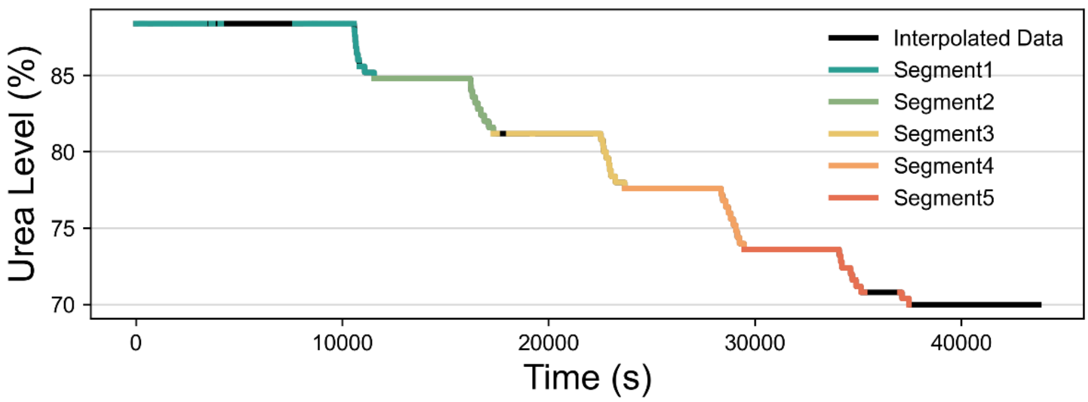

Figure 5. An Example of the Urea Consumption Segment Identification Algorithm. The black line

represents the data after signal denoising and linear interpolation. Colored lines indicate the segments identified by the algorithm.

We computed the median urea consumption rate and NOx EF for each vehicle, as shown in Figure 6. The median is chosen instead of the mean to resist possible outliers. The plot reveals no clear relationship between the two metrics. Even among vehicles compliant with the standard, urea consumption rates vary widely—from values near zero to as high as 2.2% Urea/kg-fuel. This is further supported by a Pearson correlation coefficient of 0.05 and a p-value of 0.34, which fails to reject the null hypothesis of no correlation. These findings align with Zhang et al.'s conclusions. This is expected: whether a vehicle emits at high or normal levels, its urea injection is determined by engine parameters such as speed and fuel injection rate, not by tailpipe NOx levels. Additionally, urea tank capacities vary by vehicle model and are not publicly disclosed, introducing uncertainty into this metric. Finally, the volume of daily data is still insufficient, with few observable depletion events. In summary, when daily-scale data are limited and multiple vehicle models are mixed, urea consumption rate alone is not a reliable indicator of abnormal urea use. Nevertheless, when combined with EFs, it is still possible to identify vehicles with potential tampering, offering valuable insights for further analysis.

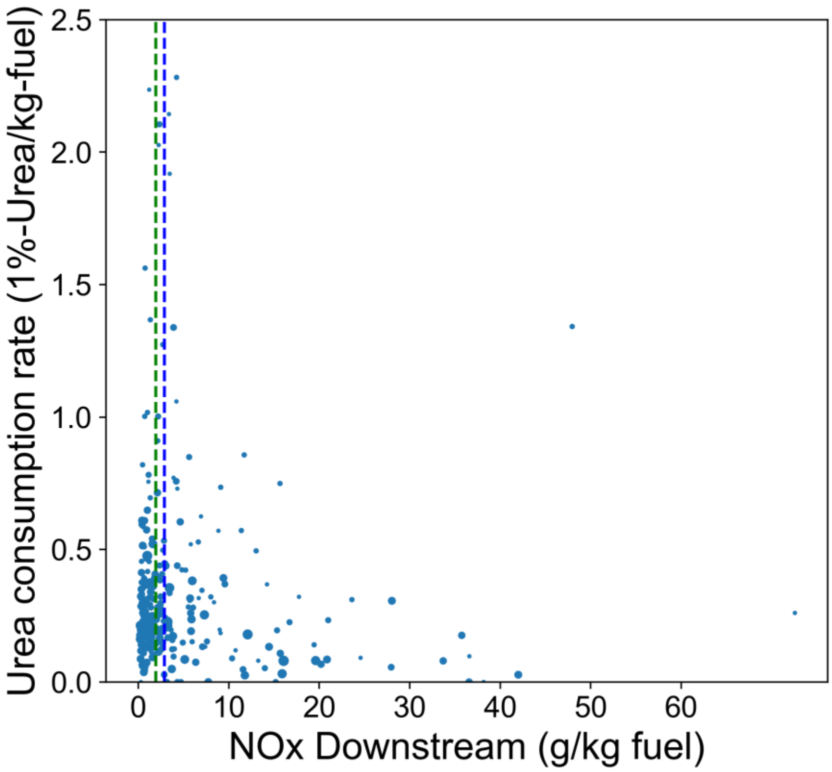

Figure 6. Fleet-level Urea Consumption Rate. The green line indicates the Diesel Engine Limits (1.94 g/kg-fuel), while the blue line represents PEMS in-use Limits (2.9 g/kg-fuel). Point size represents the number of segments detected for the corresponding vehicle.

3.3 Analysis of Urea Consumption Rates Among Vehicles of the Same Model

Since there is considerable uncertainty in comparing urea consumption rates across vehicle models, we took a further step by investigating the feasibility of this approach within vehicles of the same model. We extracted all urea consumption segments from year-scale vehicles and calculated their respective consumption rates. The resulting frequency distributions and kernel density plots are shown in figure 7. The first five vehicles, whose EFs fall below both the PEMS and engine emission limits, display similar median urea consumption rates—around 0.07% Urea/kg-fuel (0.074%, 0.069, 0.065%,0.069%, 0.071% Urea/kg-fuel,). In addition, the kernel density curve exhibits a quasi-normal distribution—dense in the middle and sparse at both ends—which is likely attributable to the varying operating conditions associated with different urea consumption segments. However, the sixth vehicle stands out, with an EF of 57.3 g/kg-fuel—far exceeding the regulatory thresholds—and a median urea consumption rate of 0.69% Urea/kg-fuel, which is markedly higher than those of vehicles with compliant emissions. Considering the number of valid segments identified of all six vehicles—151, 276, 1154, 112, 491 and 19 segments respectively—this vehicle is likely problematic. Moreover, the distorted distribution of urea consumption rate further supports this observation. The last vehicle shows a sparse distribution, characterized by both extremely low consumption segments—close to 0—and unusually high ones exceeding 1.6% urea per kg of fuel. Such a distribution appears unnatural and more likely results from a specific malfunction rather than normal urea consumption behavior.

Further investigation supports our assumption. Figure S4 show that, over an extended period, the urea level in this vehicle fluctuated around 40%, suggesting abnormal behavior. Some abnormal segments were long in duration and thus were not removed by the signal filtering algorithm. These segments were subsequently identified by the consumption-segment detection algorithm, leading to an inflated estimation of the urea consumption rate. This strongly suggests potential tampering and warrants focused investigation. Therefore, a viable approach would be to analyze vehicles of the same model to establish a baseline urea consumption rate. By comparing the number of identified urea consumption segments and their average rates against this baseline, vehicles with potential modifications or abnormally urea injection can be comprehensively identified.

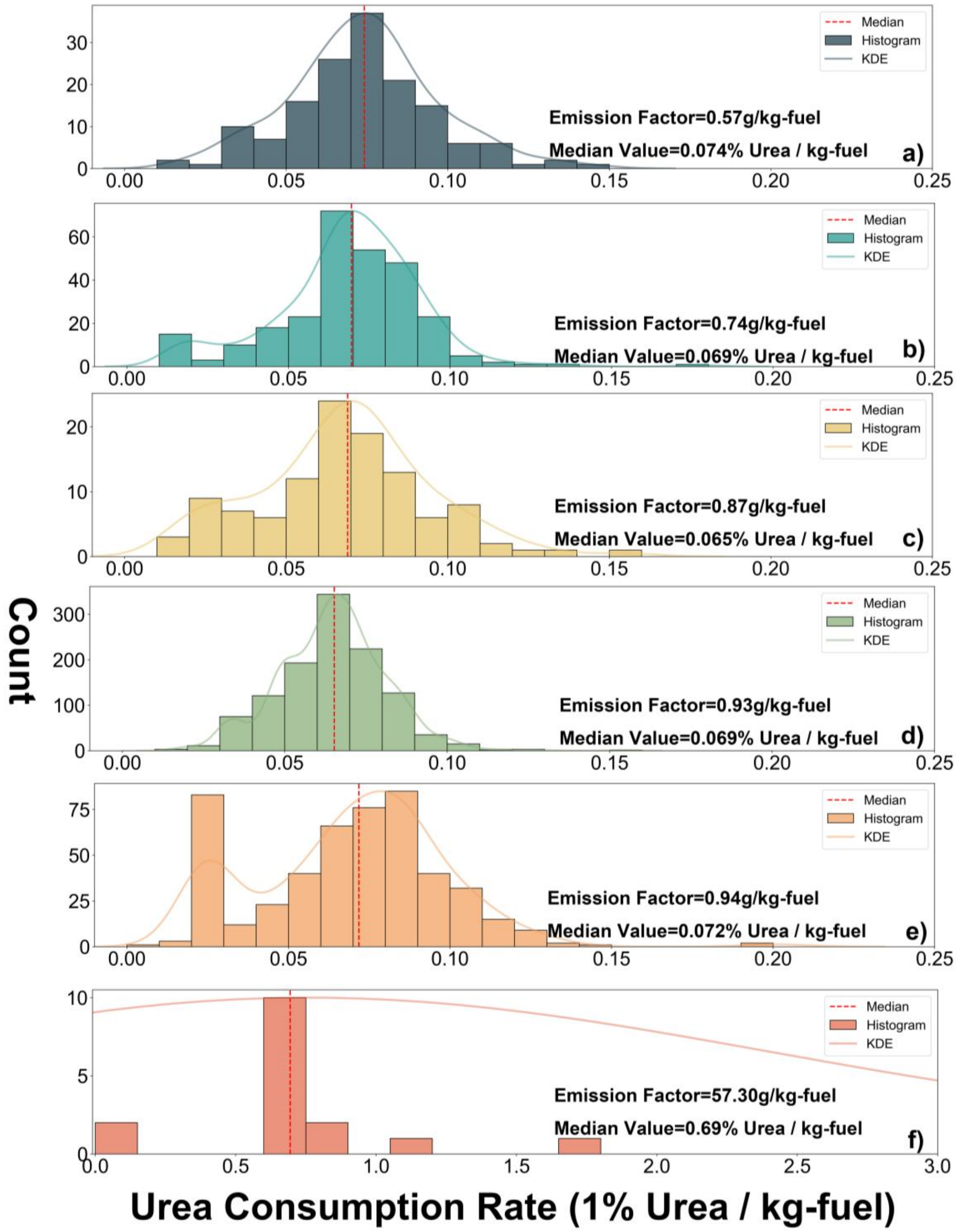

Figure 7. Distribution of Urea Consumption Rates Among Identical Vehicle Models. The red line indicates the median urea consumption rate of the corresponding vehicle, and the KDE plot (Kernel Density Estimation) illustrates the smoothed distribution.

To validate this concept, we applied the same method to 22 monthly-scale vehicles. Among them, two vehicles had no valid segments, with EFs of 35.96 g/kg-fuel and 39.45 g/kg-fuel, respectively—indicating potential tampering. For the remaining 20 vehicles, the frequency distributions and kernel density plots of their urea consumption rates are shown in the Figure S5. Most vehicles had median urea consumption rates around 0.07% Urea/kg-fuel. We used the median urea consumption rates of vehicles with EFs below both the PEMS and diesel engine limits as a baseline. According to the Kolmogorov–Smirnov test, this baseline distribution follows a normal distribution, with a p-value of 0.54711. Based on this baseline, vehicles e) and l) exhibited significantly elevated median consumption rates—0.131% and 0.139% Urea/kg-fuel, respectively—qualifying as outliers relative to the baseline. Their EFs (17.89 and 23.22 g/kg-fuel) further suggest potential abnormalities. Although an examination of their raw urea level data did not reveal clear anomalies as observed in the previous case, the results still suggest possible irregularities—such as catalyst degradation within the SCR system. This degradation could result in elevated downstream NOx concentrations, prompting the ECU to command higher urea injection rates in an attempt to neutralize the excess NOx. Such scenarios require on-site vehicle inspections to confirm. Notably, vehicle k) also exhibited an EF exceeding the standard (3.29 g/kg-fuel); however, its urea injection level displayed a healthy distribution, with a median of 0.07% Urea/kg-fuel, indicating that the urea injection system was likely functioning properly.

In summary, our current method proves applicable to vehicles of the same model and enables the identification of potentially tampered or malfunctioning vehicles. This approach can thus provide targeted inspection recommendations to regulatory authorities and serve as a methodological foundation for further studies. Moreover, as the method is purely numerical, mechanistically traceable, and computationally efficient, it holds promise for large-scale implementation.

4. Conclusion.

This study developed a segment identification algorithm for urea consumption in HDDVs, based on signal processing and moving average techniques. A method for calculating the urea consumption rate within each identified segment was also proposed. The results demonstrate that the developed method is effective and can be used to quantify fleet-level urea consumption rates. Further investigation into vehicles of the same model reveals that establishing a baseline consumption rate from well-performing vehicles, combined with information on EFs and count of usable windows, provides an efficient strategy for identifying vehicles with abnormal urea consumption behavior. The main conclusions are as follows:

1. Among the 460 vehicles analyzed, 104 had no usable data. In the remaining 356 vehicles, significant timestamp duplication was observed. However, apart from this issue, the data quality was generally acceptable, with a median availability rate of approximately 70%.
2. Within the fleet, 47% of the vehicles had EFs exceeding the PEMS in-use limits, and 58% exceeded the engine emission limits, indicating a substantial risk of excessive emissions.
3. When vehicles of various types are mixed, it is difficult to identify abnormal urea

consumption solely based on urea consumption rates and EFs derived from OBD data. However, the number of valid consumption segments identified by the segmentation algorithm serves as a potential indicator of tampering. Among the 328 daily-scale vehicles, 23 were flagged as potentially tampered—approximately 7% of the sample.

4. For vehicles of the same model, long-term time series data allowed the construction of a baseline urea consumption rate. This baseline enabled the identification of both suspected tampered and anomalous vehicles. Among 28 vehicles, the median urea consumption rate was 0.07% Urea/kg-fuel; three vehicles exhibited strong indications of tampering, and two showed signs of abnormal behavior.

At present, OBD systems have been widely deployed across China. To fully mine the potential of OBD data, it is crucial to develop tailored data mining algorithms. This study represents a step in that direction by proposing a purely numerical method for identifying urea consumption segments and calculating their urea consumption rates. The proposed algorithm is computationally efficient and holds strong potential for large-scale implementation. If broadly applied, the derived indicators could assist policymakers in identifying vehicles with abnormal urea consumption patterns, thereby enabling targeted enforcement, eliminating super-emitters, and ultimately reducing NOx emissions at their source. Nonetheless, certain limitations remain. First, both the median filter parameters used for signal denoising and the window size used in the segmentation algorithm are fixed values. While they performed well in this study, a future adaptive parameter optimization method—responsive to the characteristics of each vehicle's data—would improve robustness in large-scale platforms. Second, ground-truth validation through physical vehicle inspection is necessary to confirm the accuracy of the method and establish reliable mappings between data-driven indicators and real-world conditions.

ASSOCIATED CONTENT

Supplementary Material.

Time series errors, boundary-exceeding errors, and constant value errors (Figure S1); Two types of urea decline patterns: a) Sensitive type; b) Threshold-triggered type with a gradual decline after reaching a certain threshold; c) Threshold-triggered type with fluctuations around two discrete levels (Figure S2); Cumulative NOx mass and processed urea level for 24 vehicles with no detected change in urea level (Figure S3). Urea consumption profile of an abnormal annual-scale vehicle (Figure S4). Distribution of urea consumption rates for 20 monthly-scale vehicles. The red lines indicate the median values. KDE represents the smoothed probability density curve (Figure S5).

Notes

The authors declare no competing financial interest

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ABBREVIATIONS

HDVs, heavy-duty vehicles; HDDVs, heavy-duty diesel vehicles; OBD, On-Board Diagnostic; CFs, Consumption factors; EFs, emission factors; PEMS, Portable Emission Measurement System; SCR, Selective Catalytic Reduction; MAF, Mass Air Flow.

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